

SUKUMO, Japan

WMO: 478970

Lat: 32.9206N

Lon: 132.6950E

Elev: 36

StdP: 14.68

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 32.5	(c) 34.3	(d) 18.8	(e) 14.2	(f) 42.7	(g) 21.0	(h) 15.7	(i) 42.8	(j) 21.2	(k) 45.3	(l) 19.0	(m) 45.8	(n) 5.7	(o) 60	(p) 0.382

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 10.8	(c) 89.6	(d) 77.5	(e) 87.8	(f) 77.6	(g) 86.4	(h) 77.3	(i) 80.0	(j) 85.6	(k) 79.2	(l) 84.8	(m) 78.5	(n) 84.1	(o) 9.0	(p) 250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.4	147.7	83.5	77.5	143.2	83.0	76.7	139.5	82.6	43.7	85.5	42.8	85.1	42.0	84.3	83.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 18.5	(b) 15.7	(c) 13.6	(e) 29.4	(f) 95.2	(g) 1.5	(h) 2.2	(i) 28.3	(j) 96.8	(k) 27.4	(l) 98.1	(m) 26.6	(n) 99.3	(o) 25.5	(p) 100.9		
(a) 18.5	(b) 15.7	(c) 13.6	(e) 27.0	(f) 80.7	(g) 1.6	(h) 1.2	(i) 25.9	(j) 81.6	(k) 25.0	(l) 82.3	(m) 24.1	(n) 83.0	(o) 22.9	(p) 83.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	63.3	45.2	46.8	52.0	59.9	67.3	72.8	80.1	81.7	76.9	68.1	58.4	49.3	(6)
(7)		DBStd	13.35	4.70	5.69	5.41	5.00	3.43	3.60	2.95	2.37	3.79	4.75	5.65	5.36	(7)
(8)		HDD50	410	163	122	40	2	0	0	0	0	0	0	3	80	(8)
(9)		HDD65	2423	615	508	402	163	16	0	0	0	0	23	208	488	(9)
(10)		CDD50	5265	13	33	103	299	538	684	932	984	808	560	254	57	(10)
(11)		CDD65	1803	0	0	0	9	89	234	467	519	358	118	9	0	(11)
(12)		CDH74	14500	0	0	1	11	229	1064	4460	5570	2715	442	8	0	(12)
(13)		CDH80	3956	0	0	0	0	12	106	1269	1913	623	33	0	0	(13)
(14)	Wind	WSAvg	6.3	6.6	6.5	6.8	6.8	6.4	6.0	5.8	6.3	6.0	5.9	6.2	6.8	(14)
(15)	Precipitation	PrecAvg	80.5	2.9	3.7	5.7	6.5	7.8	11.8	9.0	9.3	10.6	6.4	4.3	2.6	(15)
(16)		PrecMax	107.0	7.1	9.4	10.6	12.2	15.8	20.8	26.6	25.2	23.5	16.7	11.7	5.1	(16)
(17)		PrecMin	54.9	0.9	1.4	3.1	3.0	3.3	4.4	1.9	0.8	2.7	1.1	1.0	0.6	(17)
(18)		PrecStd	15.0	1.7	1.9	1.9	2.4	3.0	4.5	5.6	6.3	6.0	4.4	2.4	1.3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	63.8	67.5	70.1	75.7	81.5	85.0	93.3	94.2	89.1	83.2	75.0	67.4	(19)
(20)			MCWB	57.3	60.9	59.3	63.0	64.3	74.7	76.7	76.5	75.8	73.7	66.8	59.9	(20)
(21)		2%	DB	59.2	64.0	67.2	73.1	78.2	82.7	88.9	90.5	86.8	80.5	72.7	64.2	(21)
(22)			MCWB	51.8	57.8	59.2	62.8	65.0	74.2	77.8	77.7	75.9	71.7	64.0	56.6	(22)
(23)		5%	DB	56.6	60.5	65.1	71.2	76.4	81.0	87.1	88.5	85.1	78.3	70.5	61.2	(23)
(24)			MCWB	48.9	53.2	57.0	61.7	65.6	73.3	77.6	77.9	75.3	68.9	62.4	53.4	(24)
(25)		10%	DB	54.2	57.3	62.6	69.2	74.8	79.3	85.6	87.1	83.6	76.5	68.1	58.7	(25)
(26)			MCWB	47.1	49.8	54.5	60.5	65.6	72.4	77.3	77.8	74.6	67.4	60.6	51.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	59.4	62.8	64.6	69.2	72.7	77.8	81.1	81.3	79.9	76.8	70.1	63.7	(27)
(28)			MCD B	62.5	66.2	67.7	72.3	75.9	82.0	86.9	86.9	83.9	80.7	73.2	65.8	(28)
(29)		2%	WB	54.4	59.0	61.1	66.8	71.0	76.6	79.8	80.2	78.5	74.1	66.5	57.9	(29)
(30)			MCD B	57.3	63.0	65.0	70.3	74.1	80.8	85.4	86.0	83.4	78.4	70.1	61.7	(30)
(31)		5%	WB	50.7	55.0	58.8	64.4	69.5	75.5	78.8	79.6	77.5	71.9	64.3	55.1	(31)
(32)			MCD B	54.8	59.4	63.2	68.9	73.0	79.7	84.5	85.4	82.6	76.0	68.7	59.7	(32)
(33)		10%	WB	47.9	51.1	56.3	62.4	67.6	74.1	78.0	78.9	76.4	69.6	62.2	52.5	(33)
(34)			MCD B	53.3	55.9	61.7	67.4	72.3	77.9	83.6	84.8	81.6	74.5	66.7	57.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	13.9	14.8	15.5	15.1	14.0	10.2	9.8	10.8	12.0	14.0	14.5	14.3	(35)
(36)			MCDBR	17.6	17.4	17.8	17.3	16.8	12.3	12.4	13.2	12.9	14.5	15.3	17.1	(36)
(37)			MCWBR	12.6	13.1	12.6	10.6	9.0	6.0	5.0	5.1	5.7	7.8	9.9	11.3	(37)
(38)		5% WB	MCD BR	15.0	15.2	14.5	12.7	11.0	8.5	10.3	10.7	10.3	10.9	12.3	14.0	(38)
(39)			MCWBR	13.1	13.5	13.0	10.0	8.6	5.2	4.8	4.8	5.4	7.1	9.5	11.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.405	0.448	0.507	0.532	0.541	0.554	0.545	0.522	0.492	0.459	0.420	0.395	(40)	
(41)		taud	2.170	2.073	1.925	1.917	1.964	2.021	2.086	2.159	2.199	2.194	2.224	2.211	(41)	
(42)		Ebn at Noon	249	250	245	245	244	240	241	245	247	245	243	245	(42)	
(43)		Edh at Noon	38	46	58	61	59	55	52	47	44	41	36	35	(43)	
(44)	All-Sky Solar Radiation	RadAvg	796	995	1322	1557	1642	1371	1697	1723	1380	1100	840	713	(44)	
(45)		RadStd	70	81	97	154	173	193	203	146	138	115	82	51	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		N/A	N/A	+1.55	N/A	+0.54	N/A	N/A	N/A	+137	N/A	
(47)	Regional (26 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page